

Inverse hyperbolic trig functions

Key Points

The reciprocal functions are

$$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}},$$

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

$$\operatorname{coth} x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

To use the inverses we can simply replace x with $\frac{1}{x}$ the reciprocal

Basic Skills

Q1. Write the exact value of $\operatorname{sech} 2$

Q2. Sketch the graphs stating the domain and range

a) $y = \operatorname{cosech} x$

b) $y = \operatorname{sech} x$

c) $y = \operatorname{coth} x$

Q3. Write down the logarithm versions of

a) $\operatorname{cosech}^{-1} x$

b) $\operatorname{sech}^{-1} x$

c) $\operatorname{coth}^{-1} x$

Competency Skills

Q4. Solve the following by using exponential form

a) $\operatorname{sech} x = \frac{1}{2}$ **b)** $\operatorname{cosech} x = -4$ **c)** $\operatorname{coth} x = 2$

Q5. Solve the following using the inverse functions

a) $\operatorname{sech} x = \frac{1}{2}$ **b)** $\operatorname{cosech} x = -4$ **c)** $\operatorname{coth} x = 2$

Q6. Sketch the graph of $y = 2 \operatorname{sech}(x - 1)$

Q7. Write the expression $\operatorname{coth}^2(2x^2)$ in exponential form

Q8. Prove that $\tanh^2 x = 1 - \operatorname{sech}^2 x$ by using exponential form definitions

Q9. Solve the equation $5 \operatorname{cosech} x - \operatorname{coth} x = 10$

Advanced Skills

Q10. Show that the equation $\tanh x \operatorname{cosech} x = 1$ has only one solution

Q11. Find k such that for all $|x| > k$ we have $|\operatorname{coth} x| < 2$

Q12. Evaluate $\lim_{x \rightarrow \infty} \frac{\tanh x}{\operatorname{sech} x}$

Q13. For the function $y = |\operatorname{sech}^2 x - 2^{-1}|$

a) Sketch the graph

b) Use the graph and an appropriate method to estimate the solutions to $|\operatorname{sech}^2 x - 2^{-1}| = x$

Extension

Q14. Determine which hyperbolic trig function grows the fastest towards infinity and which grows the slowest at infinity

Hint: Consider the exponential form